

CONNECTING DATA TO MODELS AND AGENTS

COURSE REFERENCE & DECISION MATRIX • RED HAT OPENSIFT AI 3.5+

1. AUTOML PREDICTIVE FOUNDATIONS (TABULAR & TIME SERIES DATA)				
Business Problem & Data Cue	Prediction Task Type	Default Optimization Metric	Key Metrics & Decision Trade-offs	Primary Output Artifacts
Churn & Fraud: 2-option outcomes (e.g. Yes/No, Approved/Denied). Data cue: 2 unique values.	Binary Classification <i>'Yes or No?'</i>	Accuracy (Switch to ROC-AUC/F1 if imbalanced)	ROC-AUC & Precision/Recall: Evaluates false positive vs. false negative trade-offs.	Leaderboard, Confusion Matrix, ROC/PR Curves, AutoGluon Predictor (S3), KServe REST API
Ticket Routing & Categorization: 3+ distinct categories (e.g. Support Tiers, Product Types).	Multiclass Classification <i>'Pick 1 from Menu'</i>	Accuracy (% of total correct predictions)	Confusion Matrix Grid: Identifies specific class-to-class misclassification patterns.	Leaderboard, Confusion Matrix, Feature Importance, Exported Jupyter Notebook (.ipynb)
Pricing & Asset Valuation: Continuous numeric estimates (e.g. House Prices, Duration).	Regression <i>'How Much / Many?'</i>	R² (R-Squared) (Scale 0.0 to 1.0; variance explained)	Feature Importance: Quantifies % impact per column (detects data leakage/noise).	Leaderboard, Feature Importance, Model Registry Entry, AutoGluon ServingRuntime
Demand & Inventory Planning: Sequential dates with Timestamp, ID, and numeric Target.	Time Series Forecasting <i>'What's Next Over Time?'</i>	MASE (Mean Absolute Scaled Error relative to naive)	Backtesting Windows: Evaluates rolling window forecast vs. observed trends.	Leaderboard, Backtesting Charts, Predictor Pipeline Artifacts, S3 Model Path
2. GENERATIVE AI GOVERNANCE & RETRIEVAL OPTIMIZATION (GENAI, MLFLOW & AUTORAG)				
Business Problem / Scenario	Platform Capability	Input / Test Requirements	Core Evaluation Metrics	Generated Artifacts & Integration
Prompt Drift & Inconsistency: Team members modify system instructions leading to unvetted outputs.	GenAI Playground & MLflow Governance	System prompts, context windows, temperature, top-p, and execution traces.	Versioned Trace Comparison: Tracks prompt parameters (v1, v2) against execution latency/cost.	Registered MLflow Prompt Templates, Execution Call-Tree Traces, Global Prompt Registry
Compliance, Legal & Healthcare: Must guarantee answers are 100% grounded in source docs (zero hallucination).	AutoRAG: Faithfulness Optimization	Document corpus (PDF, DOCX, S3) + Ground-Truth Evaluation JSON file.	Answer Faithfulness (0–1): Measures if output relies strictly on retrieved context.	Leaderboard, Sample Q&A; Comparison, 'Try This Pattern' Chat Panel, Responses API Code
Customer Support & FAQ Automation: High-volume queries where exact phrasing alignment is critical.	AutoRAG: Correctness Optimization	Evaluation JSON with question, correct_answers, and document_ids.	Answer Correctness (0–1): Measures semantic match to expected ground-truth answers.	Leaderboard, Indexing Notebook (.ipynb), Inference Notebook (.ipynb), Responses API Code
Exploratory / Knowledge Search: Broad corpora where document retrieval accuracy is the bottleneck.	AutoRAG: Retrieval Optimization	Multi-lingual docs (bge-m3 / nomic) + Vector DB (Milvus / pgvector).	Context Correctness (0–1): Measures if vector retrieval pulled the exact right chunks.	Visual Pipeline DAG, Responses API (cURL, Python, Go, Node.js), S3 Indexing Artifacts
3. DIAGNOSTIC LOGIC & SOLUTION DECISION MATRIX				
Observed Metric Diagnostic Pattern	Underlying Root Cause	Recommended Remediation Action		Execution Horizon
High Faithfulness + Low Answer Correctness (AutoRAG)	Model is strictly grounded, but retrieved chunks lack the actual ground-truth answer.	Expand S3 document corpus or switch chunking from Recursive to Hybrid (Docling) in AutoRAG.		AutoRAG Tuning
High Correctness + Low Faithfulness (AutoRAG)	Model is ignoring retrieved context and relying on innate LLM training memory (hallucination risk).	Increase top-k retrieval count or select a foundation model with stricter system instruction compliance.		RAG Pattern Refinement
High Training Accuracy + Low Test Score (AutoML)	Model is overfitting or future temporal data leaked into training feature splits.	Remove leaky columns (e.g. PassengerId) and switch preset from speed to balanced quality.		AutoML Feature Hygiene
Auto Capabilities Max Out On Complex Cases (Escalation Gate)	Domain complexity requires task-specific behavior, synthetic edge data, or fine-tuning.	Escalate to Course 2: Ingest with Docling, generate Q&A; via SDG Hub, fine-tune via Training Hub.		Model Customization